

CH Detected-Copy1

July 20, 2026

```
[1]: # -----  
# Standard libraries  
# -----  
import numpy as np  
import pandas as pd  
  
# -----  
# Astropy / Coordinates / Units  
# -----  
import astropy.units as u  
from astropy.coordinates import SkyCoord  
  
# -----  
# SunPy (maps, solar constants, and data access)  
# -----  
import sunpy.map  
from sunpy.sun import constants as sun_const  
from sunpy.net import Fido  
from sunpy.net import attrs as a  
  
# -----  
# Image processing  
# -----  
from skimage.measure import label, regionprops  
from scipy.ndimage import gaussian_filter  
from aiapy.calibrate import register  
  
# -----  
# Plotting and visualization  
# -----  
import matplotlib.pyplot as plt  
from matplotlib.colors import Normalize  
from matplotlib.lines import Line2D  
  
# -----  
# Enable interactive plotting  
# -----
```

```
plt.ion()
```

```
[1]: <contextlib.ExitStack at 0x73759689b5f0>
```

```
[2]: AIA_DATA = sunpy.map.Map("/home/bhairab/Downloads/fitsfile/14/23/fits file/aia.  
    ↪lev1.193A_2025_09_14T23_59_04.84Z.image_lev1.fits")  
HMI_DATA = sunpy.map.Map("/home/bhairab/Downloads/fitsfile/14/23/fits file/hmi.  
    ↪m_45s.2025.09.15_00_00_00_TAI.magnetogram.fits")
```

```
[3]: # Create SunPy Map objects from the downloaded FITS files  
# aia_193: Representative of the hot solar corona  
aia193 = sunpy.map.Map(AIA_DATA)  
  
# hmi6173: Representative of the magnetic photosphere  
hmi6173 = sunpy.map.Map(HMI_DATA)
```

```
[4]: # -----  
# ALIGNMENT: Reprojecting AIA to match HMI coordinates  
# -----  
aia_reprojected = aia193.reproject_to(hmi6173.wcs)  
aia_reprojected.nickname = 'AIA 193 Å (Reprojected)'
```

2026-07-19 10:04:54 - reproject.common - INFO: Calling _reproject_full in non-dask mode

```
[5]: # =====  
# PROCESSING: Normalize AIA 193 Å by Exposure Time  
# =====  
# Use exposure time from the ORIGINAL AIA map (not reprojected)  
aia_exptime = aia193.exposure_time.to_value() # seconds  
  
# Normalize reprojected map using original exposure time  
aia_norm = aia_reprojected / aia_exptime # DN/s  
  
# =====  
# FIGURE SETUP  
# =====  
fig = plt.figure(figsize=(12, 5))  
  
# =====  
# PANEL 1: AIA 193 Å (Exposure-Normalized, AIA Colormap)  
# =====  
ax1 = fig.add_subplot(121, projection=aia_reprojected)  
  
im1 = aia_norm.plot(  
    axes=ax1,
```

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        cmap="sdoaia193",
        clip_interval=(1, 99.9) * u.percent
    )

    ax1.set_title(
        "AIA 193 Å (DN/s) 2025 Sept. 10",
        fontsize=14
    )

    cb1 = fig.colorbar(
        im1,
        ax=ax1,
        orientation='vertical',
        fraction=0.046,
        pad=0.04
    )
    cb1.set_label("Intensity (DN/s)")

    ax1.invert_xaxis()
    ax1.invert_yaxis()

    # =====
    # PANEL 2: HMI 6173 Å (LOS Magnetic Field, Grayscale)
    # =====
    ax2 = fig.add_subplot(122, projection=hmi6173)

    im2 = hmi6173.plot(
        axes=ax2,
        cmap="gray",
        norm=Normalize(-500, 500)
    )

    ax2.set_title(
        "HMI 6173 Å - 2025 Sept. 10",
        fontsize=14
    )

    cb2 = fig.colorbar(
        im2,
        ax=ax2,
        orientation='vertical',
        fraction=0.046,
        pad=0.04
    )
    cb2.set_label("B$_{LOS}$ [G]")

```

```

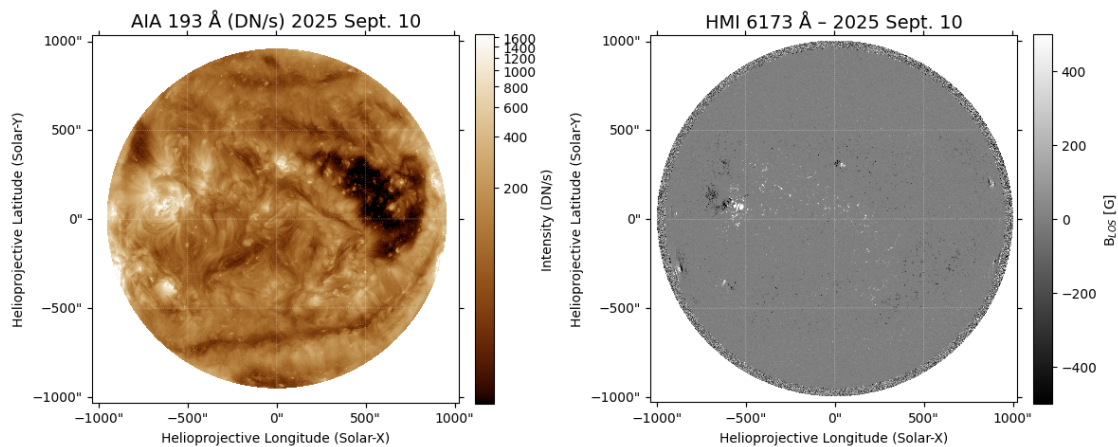
ax2.invert_xaxis()
ax2.invert_yaxis()

# =====
# FINAL LAYOUT & EXPORT
# =====
plt.tight_layout(pad=1.2)

#plt.savefig(    "/home/bhairab/Documents/project/reproject.png",    dpi=600,
    ↪ bbox_inches="tight")

plt.show()

```



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[6]: # -----
# Define corners in the AIA coordinate frame
# -----
aia_bottom_left = SkyCoord(50 * u.arcsec, -350 * u.arcsec,
                           frame=aia_reprojected.coordinate_frame)
aia_top_right    = SkyCoord(900 * u.arcsec, 450 * u.arcsec,
                           frame=aia_reprojected.coordinate_frame)

# -----
# Create figure
# -----
fig = plt.figure(figsize=(12, 6))

# ===== AIA Plot =====
ax1 = fig.add_subplot(121, projection=aia_reprojected.wcs)
im1 = aia_reprojected.plot(
    axes=ax1,

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        clip_interval=(1, 99.9) * u.percent
    )

    aia_reprojected.draw_quadrangle(
        bottom_left=aia_bottom_left,
        top_right=aia_top_right,
        axes=ax1,
        edgecolor="black",
        linestyle="--",
        linewidth=2
    )

    ax1.set_title("AIA 193 Å 2025 Sept. 08 06:00:34", fontsize=14)

    ax1.coords[0].set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
    ax1.coords[1].set_xlabel("Helioprojective Latitude [arcsec]", fontsize=13)

    # Enforce uniform label rendering (Astropy-safe)
    ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
    ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

    ax1.tick_params(labelsize=12)
    ax1.invert_xaxis()
    ax1.invert_yaxis()

    cbar1 = fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.02)
    cbar1.set_label("Intensity [DN]", fontsize=13)
    cbar1.ax.tick_params(labelsize=10)

    ax1.text(
        0.02, 0.95, "a",
        transform=ax1.transAxes,
        va="top", ha="left",
        fontsize=15
    )

    # ===== HMI Plot =====
    ax2 = fig.add_subplot(122, projection=hmi6173.wcs)
    im2 = hmi6173.plot(
        axes=ax2,
        cmap="grey",
        norm=Normalize(-500, 500)
    )

    hmi6173.draw_quadrangle(
        bottom_left=aia_bottom_left,
        top_right=aia_top_right,

```

```

        axes=ax2,
        edgecolor="black",
        linestyle="--",
        linewidth=2
    )

ax2.set_title("HMI 6173 Å 2025 Sept. 08 06:00:34", fontsize=14)

ax2.coords[0].set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.coords[1].set_xlabel("Helioprojective Latitude [arcsec]", fontsize=13)

# Enforce uniform label rendering (Astropy-safe)
ax2.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

ax2.tick_params(labelsize=12)
ax2.invert_xaxis()
ax2.invert_yaxis()

cbar2 = fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.02)
cbar2.set_label(r"$B_{\mathrm{LOS}} \setminus \mathrm{[G]}$", fontsize=13)
cbar2.ax.tick_params(labelsize=10)

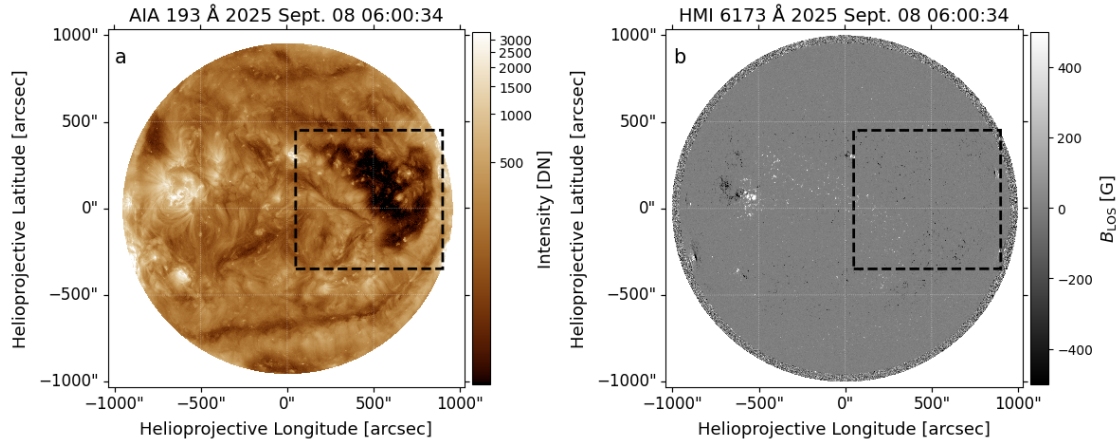
ax2.text(
    0.02, 0.95, "b",
    transform=ax2.transAxes,
    va="top", ha="left",
    fontsize=15
)

# -----
# Layout + Save
# -----
plt.tight_layout(pad=1.2)

#plt.savefig(    "/home/bhairab/Documents/project/chbox.png",    dpi=600,    □
↪ bbox_inches="tight")

plt.show()

```



```
[7]: # -----
# Global ApJ-safe style (ONE place)
# -----
plt.rcParams.update({
    "font.family": "serif",
    "mathtext.fontset": "cm",
    "xtick.direction": "in",
    "ytick.direction": "in",
})

# -----
# Create submaps from defined region
# -----
aia_sub = aia_reprojected.submap(
    bottom_left=aia_bottom_left,
    top_right=aia_top_right
)
hmi_sub1 = hmi6173.submap(
    bottom_left=aia_bottom_left,
    top_right=aia_top_right
)

# -----
# Create figure
# -----
fig = plt.figure(figsize=(12, 6))

# ===== AIA submap =====
ax1 = fig.add_subplot(121, projection=aia_sub.wcs)
im1 = aia_sub.plot(
    axes=ax1,
```

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        clip_interval=(1, 99.9) * u.percent
    )

    ax1.set_title("AIA 193 Å 2025 Sept. 08 06:00:34", fontsize=14)

    ax1.coords[0].set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
    ax1.coords[1].set_xlabel("Helioprojective Latitude [arcsec]", fontsize=13)

    # Enforce uniform label rendering (Astropy-safe)
    ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
    ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

    ax1.tick_params(labelsize=12)
    ax1.invert_xaxis()
    ax1.invert_yaxis()

    cbar1 = fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.02)
    cbar1.set_label("Intensity [DN]", fontsize=13)
    cbar1.ax.tick_params(labelsize=10)

    ax1.text(
        0.02, 0.95, "a",
        transform=ax1.transAxes,
        va="top", ha="left",
        fontsize=15
    )

    # ===== HMI submap =====
    ax2 = fig.add_subplot(122, projection=hmi_sub1.wcs)
    im2 = hmi_sub1.plot(
        axes=ax2,
        cmap="grey",
        norm=Normalize(-500, 500)
    )

    ax2.set_title("HMI 6173 Å 2025 Sept. 08 06:00:34", fontsize=14)

    ax2.coords[0].set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
    ax2.coords[1].set_xlabel("Helioprojective Latitude [arcsec]", fontsize=13)

    # Enforce uniform label rendering (Astropy-safe)
    ax2.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
    ax2.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

    ax2.tick_params(labelsize=12)
    ax2.invert_xaxis()
    ax2.invert_yaxis()

```



```

cbar2 = fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.02)
cbar2.set_label(r"$B_{\mathrm{LOS}} \setminus \mathrm{[G]}$", fontsize=13)
cbar2.ax.tick_params(labelsize=10)

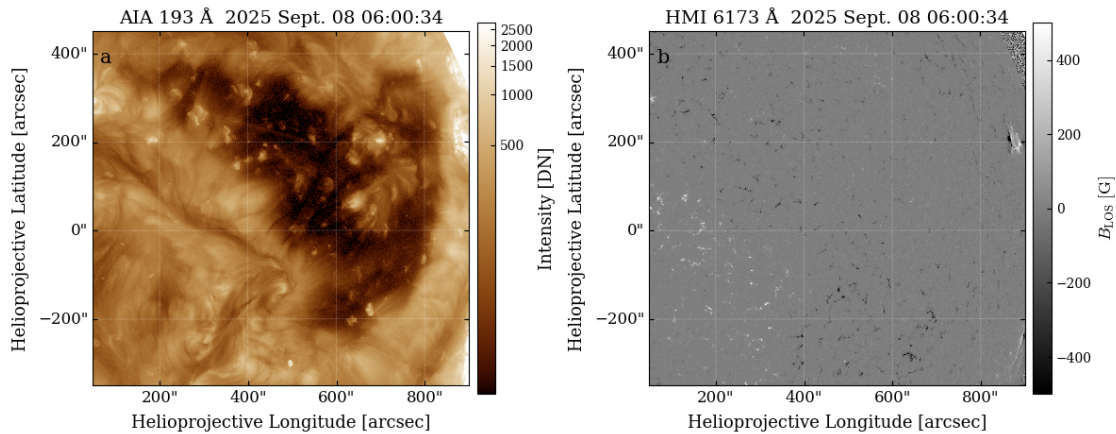
ax2.text(
    0.02, 0.95, "b",
    transform=ax2.transAxes,
    va="top", ha="left",
    fontsize=15
)

# -----
# Final layout and show/save
# -----
plt.tight_layout(pad=1.0)

#plt.savefig(    "/home/bhairab/Documents/project/chonly.png",    dpi=600,    ↵
    ↳bbox_inches="tight")

plt.show()

```



```

[8]: # Global ApJ-uniform style (SAFE) - same as above
# -----
plt.rcParams.update({
    "font.family": "serif",
    "mathtext.fontset": "cm",
    "xtick.direction": "in",
    "ytick.direction": "in",
})

```

```

# -----
# Segment AIA dark regions + smooth
# -----
threshold = 100
segmented = aia_sub.data < threshold
smoothed_mask = gaussian_filter(segmented.astype(float), sigma=2) > 0.5

# Label regions (use AIA intensity image for region finding)
labeled = label(smoothed_mask)
regions = regionprops(labeled, intensity_image=aia_sub.data)
regions = sorted(regions, key=lambda r: r.area, reverse=True)

# Safety: handle empty detection
if len(regions) == 0:
    raise ValueError("No regions found. Try adjusting threshold/sigma.")

# Keep only largest region
largest_region = regions[0]
single_mask = (labeled == largest_region.label)

# Region centroid (pixel)
cy, cx = largest_region.centroid # (row, col)

# -----
# Plot (UNIFORM with the earlier figure: GridSpec + per-axis colorbars)
# -----
fig = plt.figure(figsize=(13, 6))
gs = fig.add_gridspec(1, 2)

# ===== AIA panel =====
ax1 = fig.add_subplot(gs[0, 0], projection=aia_sub.wcs)
im1 = aia_sub.plot(axes=ax1, clip_interval=(1, 99.9) * u.percent)
ax1.contour(single_mask, levels=[0.5], colors="red", origin="lower")

ax1.set_title("AIA 193 Å 2025 Sept. 08 06:00:34", fontsize=14)

ax1.coords[0].set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.coords[1].set_xlabel("Helioprojective Latitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

# WCS tick labels (PDF consistency)
ax1.coords[0].set_ticklabel(size=12)
ax1.coords[1].set_ticklabel(size=12)

# label at centroid (bold like your other labels)
ax1.text(cx, cy, "0", color="white",

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        ha="center", va="center", fontsize=15, fontweight="bold")

# panel label
ax1.text(0.02, 0.95, "a",
        transform=ax1.transAxes, va="top", ha="left",
        fontsize=15, color="black")

ax1.invert_xaxis()
ax1.invert_yaxis()

cbar1 = fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.02)
cbar1.set_label("Intensity [DN]", fontsize=13)
cbar1.ax.tick_params(labelsize=10)

# ===== HMI panel =====
ax2 = fig.add_subplot(gs[0, 1], projection=hmi_sub1.wcs)
im2 = hmi_sub1.plot(axes=ax2, cmap="grey", norm=Normalize(-500, 500))
ax2.contour(single_mask, levels=[0.5], colors="red", origin="lower")

ax2.set_title("HMI 6173 Å 2025 Sept. 08 06:00:34", fontsize=14)

ax2.coords[0].set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.coords[1].set_xlabel("Helioprojective Latitude [arcsec]", fontsize=13)
ax2.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

# WCS tick labels (PDF consistency)
ax2.coords[0].set_ticklabel(size=12)
ax2.coords[1].set_ticklabel(size=12)

# same centroid label
ax2.text(cx, cy, "0", color="white",
        ha="center", va="center", fontsize=15)

# panel label
ax2.text(0.02, 0.95, "b",
        transform=ax2.transAxes, va="top", ha="left",
        fontsize=15, color="black")

ax2.invert_xaxis()
ax2.invert_yaxis()

cbar2 = fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.02)
cbar2.set_label(r"$B_{\mathrm{LOS}} \setminus \mathrm{[G]}$", fontsize=13)
cbar2.ax.tick_params(labelsize=10)

# -----

```

```

# IMPORTANT: no tight_layout for WCSAxes + PDF
# -----

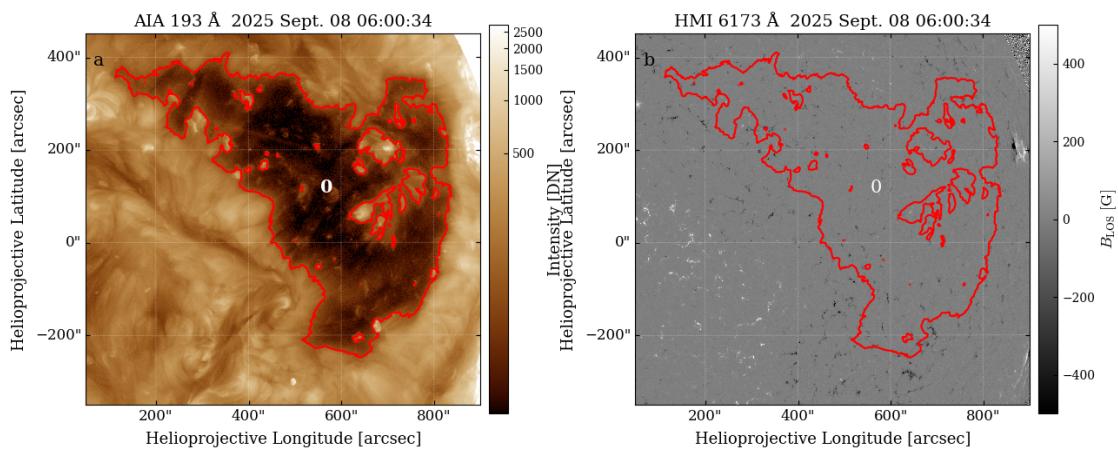
# -----
# Layout + Save
# -----

plt.subplots_adjust(left=0.07, right=0.93, bottom=0.10, top=0.90, wspace=0.30)

#plt.savefig(      "/home/bhairab/Documents/project/ch_boundary.png",      dpi=600,
#               ↪   bbox_inches="tight")

plt.show()

```



```

[9]: # Create mask from largest region
bbox = regions[0].bbox
mask = np.ones_like(hmi_sub1.data, dtype=bool)
mask[bbox[0]:bbox[2],bbox[1]:bbox[3]] = ~regions[0].image

# Create masked SunPy map
hmi_masked = sunpy.map.Map((hmi_sub1.data, hmi_sub1.meta), mask=mask)

# Extract unmasked data
unmasked_data = hmi_masked.data[~hmi_masked.mask]

# Create figure with two subplots
fig = plt.figure(figsize=(18, 6))

# ===== Masked HMI Map =====
ax1 = fig.add_subplot(121, projection=hmi_masked)
im = hmi_masked.plot(axes=ax1, cmap="grey", norm=Normalize(-500, 500))

```

```

ax1.set_title("HMI 6173 Å 2025 Sept. 08 06:00:34", fontsize=14)
ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)
ax1.tick_params(labelsize=12, direction="in")

ax1.invert_xaxis()
ax1.invert_yaxis()

# Colorbar EXACTLY like the AIA+HMI figure (fraction + pad only)
cbar1 = fig.colorbar(im, ax=ax1, fraction=0.046, pad=0.02)
cbar1.set_label(r"$B_{\mathrm{LOS}} \setminus \mathrm{[G]}$", fontsize=13)
cbar1.ax.tick_params(labelsize=10)

# Panel label
ax1.text(0.02, 0.95, "c",
        transform=ax1.transAxes,
        va="top", ha="left",
        fontsize=15)

# ===== Histogram =====
ax2 = fig.add_subplot(122)
ax2.hist(unmasked_data, bins=50, histtype="step", color="blue", linewidth=1.5)

ax2.set_yscale("log")
ax2.set_xlabel(r"$B_{\mathrm{LOS}} \setminus \mathrm{[G]}$", fontsize=13)
ax2.set_ylabel("Number of pixels", fontsize=13)
ax2.set_title(r"Distribution of $B_{\mathrm{LOS}}$", fontsize=14)
ax2.tick_params(labelsize=12, direction="in")

# Annotations
ax2.text(0.55, 0.95, r"$S = 0.37$",
        transform=ax2.transAxes,
        va="top", ha="left",
        fontsize=14)

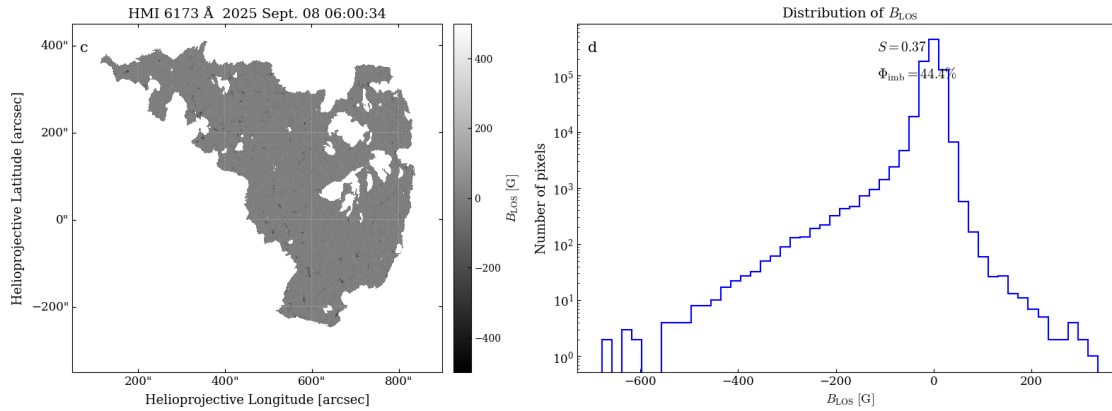
ax2.text(0.55, 0.88, r"$\Phi_{\mathrm{imb}} = 44.4\%$",
        transform=ax2.transAxes,
        va="top", ha="left",
        fontsize=14)

# Panel label
ax2.text(0.02, 0.95, "d",
        transform=ax2.transAxes,
        va="top", ha="left",
        fontsize=15)

# Layout + Save

```

```
# -----
plt.tight_layout(pad=1.2)
#plt.savefig(      "/home/bhairab/Documents/project/Coronal_map_histogram1.png",
    ↪ dpi=600,      bbox_inches="tight")
plt.show()
```



```
[10]: # ---- AIA: average intensity inside CH mask (DN)
aia_avg_intensity = np.mean(aia_sub.data[single_mask])

# ---- Number of pixels in CH
N_pixels = single_mask.sum() # Only CH pixels!

# ---- Pixel scale (arcsec / pixel)
scale_x = hmi_masked.scale[0].to_value(u.arcsec / u.pixel)
scale_y = hmi_masked.scale[1].to_value(u.arcsec / u.pixel)

# ---- Constants
arcsec_to_rad = (1 * u.arcsec).to_value(u.rad)
R_sun_cm = sun_const.radius.to_value(u.cm)

# ---- Observer distance (cm) - use actual Sun-observer distance for THIS
    ↪ observation,
# NOT the solar radius. This was the bug causing flux to be ~4.6e4x too small.
dsun_cm = hmi_masked.dsun.to_value(u.cm)

# ---- Pixel area (cm2)
pixel_area_cm2 = (scale_x * arcsec_to_rad * dsun_cm) * \
    (scale_y * arcsec_to_rad * dsun_cm)

# ---- Coronal hole area
ch_area_cm2 = N_pixels * pixel_area_cm2
ch_area_m2 = ch_area_cm2 * 1e-4
```

```

ch_area_Mm2 = ch_area_cm2 * 1e-16

# ---- Total unsigned magnetic flux (Mx)
total_unsigned_flux = np.sum(
    np.abs(hmi_sub1.data[single_mask]) * pixel_area_cm2
)

# ---- Print results
#print(f"Observation datetime      = {observation_datetime}")
print("=====  

Coronal Hole Properties =====")
print(f"Average AIA intensity      = {aia_avg_intensity:.2f} DN")
print(f"Total unsigned magnetic flux = {total_unsigned_flux:.2e} Mx")
print(f"Coronal Hole area              = {ch_area_m2:.2e} m^2")
print(f"Coronal Hole area              = {ch_area_Mm2:.2f} Mm^2")

=====  

Coronal Hole Properties =====
Average AIA intensity      = 52.61 DN
Total unsigned magnetic flux = 1.29e+22 Mx
Coronal Hole area          = 1.07e+17 m^2
Coronal Hole area          = 106708.82 Mm^2

```